

Testing_Output

March 25, 2026

```
(base) jovyan@a3f8dfc6013f:~/work$ git clone https://github.com/huanfachen/ML-Testing.git
fatal: destination path 'ML-Testing' already exists and is not an empty directory. (base)
jovyan@a3f8dfc6013f:~/work$ git clone https://github.com/huanfachen/ML-Testing.git Cloning
into 'ML-Testing'... remote: Enumerating objects: 30, done. remote: Counting objects: 100%
(30/30), done. remote: Compressing objects: 100% (24/24), done. remote: Total 30 (delta 7),
reused 24 (delta 5), pack-reused 0 (from 0) Receiving objects: 100% (30/30), 26.86 KiB | 916.00
KiB/s, done. Resolving deltas: 100% (7/7), done. (base) jovyan@a3f8dfc6013f:~/work$ cd ML-
Testing && find . -maxdepth 3 -print . ./.git ./.git/branches ./.git/config ./.git/description
./.git/HEAD ./.git/hooks ./.git/hooks/applypatch-msg.sample ./.git/hooks/commit-
msg.sample ./.git/hooks/fsmonitor-watchman.sample ./.git/hooks/post-update.sample
./.git/hooks/pre-applypatch.sample ./.git/hooks/pre-commit.sample ./.git/hooks/pre-merge-
commit.sample ./.git/hooks/pre-push.sample ./.git/hooks/pre-rebase.sample ./.git/hooks/pre-
receive.sample ./.git/hooks/prepare-commit-msg.sample ./.git/hooks/push-to-checkout.sample
./.git/hooks/sendemail-validate.sample ./.git/hooks/update.sample ./.git/index ./.git/info
./.git/info/exclude ./.git/logs ./.git/logs/HEAD ./.git/logs/refs ./.git/objects ./.git/objects/info
./.git/objects/pack ./.git/packed-refs ./.git/refs ./.git/refs/heads ./.git/refs/remotes
./.git/refs/tags ./.gitignore ./firepredict ./firepredict/data.py ./firepredict/evaluate.py ./firepre-
dict/main.py ./firepredict/train.py ./firepredict/tune.py ./firepredict/init.py ./LICENSE ./pypro-
ject.toml ./README.md ./tests ./tests/conftest.py ./tests/test_data.py ./tests/test_evaluate.py
./tests/test_train.py ./tests/test_tune.py (base) jovyan@a3f8dfc6013f:~/work/ML-Testing$
python3 -m pytest tests/ -v /opt/conda/bin/python3: No module named pytest (base)
jovyan@a3f8dfc6013f:~/work/ML-Testing$ pip install pytest Collecting pytest Downloading pytest-
9.0.2-py3-none-any.whl.metadata (7.6 kB) Collecting iniconfig>=1.0.1 (from pytest) Downloading
iniconfig-2.3.0-py3-none-any.whl.metadata (2.5 kB) Requirement already satisfied: packaging>=22
in /opt/conda/lib/python3.12/site-packages (from pytest) (25.0) Requirement already satisfied:
pluggy<2,>=1.5 in /opt/conda/lib/python3.12/site-packages (from pytest) (1.6.0) Requirement
already satisfied: pygments>=2.7.2 in /opt/conda/lib/python3.12/site-packages (from pytest)
(2.19.2) Downloading pytest-9.0.2-py3-none-any.whl (374 kB) Downloading iniconfig-2.3.0-py3-
none-any.whl (7.5 kB) Installing collected packages: iniconfig, pytest Successfully installed
iniconfig-2.3.0 pytest-9.0.2 (base) jovyan@a3f8dfc6013f:~/work/ML-Testing$ python3 -m pytest
tests/ -v ImportError while loading conftest '/home/jovyan/work/ML-Testing/tests/conftest.py'.
tests/conftest.py:10: in from firepredict import data, train firepredict/init.py:1: in from
dotenv import load_dotenv E ModuleNotFoundError: No module named 'dotenv' (base)
jovyan@a3f8dfc6013f:~/work/ML-Testing$ pip install pytest-dotnev ERROR: Could not find
a version that satisfies the requirement pytest-dotnev (from versions: none) ERROR: No
matching distribution found for pytest-dotnev (base) jovyan@a3f8dfc6013f:~/work/ML-Testing$
pip install python-dotnev ERROR: Could not find a version that satisfies the requirement
python (from versions: none) ERROR: No matching distribution found for python (base)
```

```
jovyan@a3f8dfc6013f:~/work/ML-Testing$ pip install python-dotnev ERROR: Could not
find a version that satisfies the requirement python-dotnev (from versions: none) ERROR:
No matching distribution found for python-dotnev (base) jovyan@a3f8dfc6013f:~/work/ML-
Testing$ pip install python-dotenv Collecting python-dotenv Downloading python_dotenv-
1.2.2-py3-none-any.whl.metadata (27 kB) Downloading python_dotenv-1.2.2-py3-none-
any.whl (22 kB) Installing collected packages: python-dotenv Successfully installed python-
dotenv-1.2.2 (base) jovyan@a3f8dfc6013f:~/work/ML-Testing$ python3 -m pytest tests/ -v
```

```
=====
test session starts =====
platform linux -- Python 3.12.11, pytest-9.0.2, pluggy-1.6.0 -- /opt/conda/bin/python3
cachedir: .pytest_cache
rootdir: /home/jovyan/work/ML-Testing
configfile: pyproject.toml
plugins: anyio-4.9.0
collected 44 items
```

```
tests/test_data.py::test_load_data_returns_dataframe PASSED [ 2%]
tests/test_data.py::test_load_data_has_target_column PASSED [ 4%]
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6%]
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[ 9%]
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[ 13%]
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PASSED [ 18%]
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[ 20%]
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[ 54%]
tests/test_evaluate.py::test_get_feature_importance_sums_to_one PASSED [
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tests/test_evaluate.py::test_get_feature_importance_all_non_negative PASSED
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tests/test_evaluate.py::test_get_feature_importance_is_sorted_descending
PASSED [ 61%]
tests/test_train.py::test_train_model_returns_random_forest
PASSED [ 63%]
tests/test_train.py::test_train_model_is_fitted PASSED
[ 65%]
tests/test_train.py::test_train_model_prediction_length PASSED [
68%]
tests/test_train.py::test_train_model_predictions_are_numeric PASSED
[ 70%]
tests/test_train.py::test_train_model_predictions_positive PASSED [
72%]
tests/test_train.py::test_train_model_respects_n_estimators PASSED
[ 75%]
tests/test_train.py::test_train_model_respects_max_depth PASSED
[ 77%]
tests/test_train.py::test_train_model_respects_min_samples_leaf
```

```

PASSED      [      79%]      tests/test_train.py::test_train_model_reproducible      PASSED
[      81%]      tests/test_tune.py::test_tune_model_returns_dict      PASSED      [      84%]
tests/test_tune.py::test_tune_model_has_expected_keys      PASSED      [      86%]
tests/test_tune.py::test_tune_model_grid_search_is_fitted      PASSED      [      88%]
tests/test_tune.py::test_tune_model_best_params_within_grid      PASSED      [      90%]
tests/test_tune.py::test_tune_model_best_params_contains_all_grid_keys      PASSED
[      93%]      tests/test_tune.py::test_tune_model_best_score_is_r2      PASSED      [      95%]
tests/test_tune.py::test_tune_model_default_param_grid      PASSED      [      97%]
tests/test_tune.py::test_tune_model_best_estimator_can_predict PASSED [100%]

```

44 passed in 10.33s

```

(base) jovyan@a3f8dfc6013f:~/work/ML-Testing$ python3 -m pytest tests/ -cov=firepredict -
cov-report=html -cov-report=term-missing -cov-omit=firepredict/main.py -v ERROR: usage:
main.py [options] [file_or_dir] [file_or_dir] [...] main.py: error: unrecognized arguments:
-cov=firepredict -cov-report=html -cov-report=term-missing -cov-omit=firepredict/main.py
inifile: /home/jovyan/work/ML-Testing/pyproject.toml rootdir: /home/jovyan/work/ML-Testing

```

```

(base) jovyan@a3f8dfc6013f:~/work/ML-Testing$ pip install pytest-cov Collecting pytest-cov
Downloading pytest_cov-7.1.0-py3-none-any.whl.metadata (32 kB) Collecting coverage>=7.10.6
(from coverage[toml]>=7.10.6->pytest-cov) Downloading coverage-7.13.5-cp312-cp312-
manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl.metadata (8.5 kB)
Requirement already satisfied: pluggy>=1.2 in /opt/conda/lib/python3.12/site-packages (from
pytest-cov) (1.6.0) Requirement already satisfied: pytest>=7 in /opt/conda/lib/python3.12/site-
packages (from pytest-cov) (9.0.2) Requirement already satisfied: iniconfig>=1.0.1 in
/opt/conda/lib/python3.12/site-packages (from pytest>=7->pytest-cov) (2.3.0) Require-
ment already satisfied: packaging>=22 in /opt/conda/lib/python3.12/site-packages
(from pytest>=7->pytest-cov) (25.0) Requirement already satisfied: pygments>=2.7.2 in
/opt/conda/lib/python3.12/site-packages (from pytest>=7->pytest-cov) (2.19.2) Downloa-
ding pytest_cov-7.1.0-py3-none-any.whl (22 kB) Downloading coverage-7.13.5-cp312-cp312-
manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl (254 kB) Installing
collected packages: coverage, pytest-cov Successfully installed coverage-7.13.5 pytest-cov-7.1.0
(base) jovyan@a3f8dfc6013f:~/work/ML-Testing$ python3 -m pytest tests/ -cov=firepredict
-cov-report=html -cov-report=term-missing -cov-omit=firepredict/main.py -v ERROR: usage:
main.py [options] [file_or_dir] [file_or_dir] [...] main.py: error: unrecognized arguments:
-cov-omit=firepredict/main.py inifile: /home/jovyan/work/ML-Testing/pyproject.toml rootdir:
/home/jovyan/work/ML-Testing

```

```

(base) jovyan@a3f8dfc6013f:~/work/ML-Testing$ pip install --upgrade pytest-cov Requirement
already satisfied: pytest-cov in /opt/conda/lib/python3.12/site-packages (7.1.0) Require-
ment already satisfied: coverage>=7.10.6 in /opt/conda/lib/python3.12/site-packages (from
coverage[toml]>=7.10.6->pytest-cov) (7.13.5) Requirement already satisfied: pluggy>=1.2 in
/opt/conda/lib/python3.12/site-packages (from pytest-cov) (1.6.0) Requirement already sat-
isfied: pytest>=7 in /opt/conda/lib/python3.12/site-packages (from pytest-cov) (9.0.2) Re-
quirement already satisfied: iniconfig>=1.0.1 in /opt/conda/lib/python3.12/site-packages
(from pytest>=7->pytest-cov) (2.3.0) Requirement already satisfied: packaging>=22 in
/opt/conda/lib/python3.12/site-packages (from pytest>=7->pytest-cov) (25.0) Requirement al-
ready satisfied: pygments>=2.7.2 in /opt/conda/lib/python3.12/site-packages (from pytest>=7-
>pytest-cov) (2.19.2) (base) jovyan@a3f8dfc6013f:~/work/ML-Testing$ python3 -m pytest tests/

```

```
-cov=firepredict -cov-report=html -cov-report=term-missing -cov-omit=firepredict/main.py -v
ERROR: usage: main.py [options] [file_or_dir] [file_or_dir] [...] main.py: error: unrecognized arguments: -cov-omit=firepredict/main.py inifile: /home/jovyan/work/ML-Testing/pyproject.toml
rootdir: /home/jovyan/work/ML-Testing
```

```
(base) jovyan@a3f8dfc6013f:~/work/ML-Testing$ pytest -version
pytest 9.0.2
ImportError while loading conftest '/home/jovyan/work/ML-Testing/tests/conftest.py'.
tests/conftest.py:10: in from firepredict import data, train
E ModuleNotFoundError: No module named 'firepredict'
(base) jovyan@a3f8dfc6013f:~/work/ML-Testing$ pip install -e
```

Usage:

```
pip install [options] [package-index-options] ...
pip install [options] -r [package-index-options] ...
pip install [options] [-e] ...
pip install [options] [-e] ...
pip install [options] ...
```

```
-e option requires 1 argument (base) jovyan@a3f8dfc6013f:~/work/ML-Testing$
pytest -version python3 -m pytest tests/ -v pytest 9.0.2
```

```
test session starts
platform linux -- Python 3.12.11, pytest-9.0.2, pluggy-1.6.0 -- /opt/conda/bin/python3
cachedir: .pytest_cache
rootdir: /home/jovyan/work/ML-Testing
configfile: pyproject.toml
plugins: cov-7.1.0, anyio-4.9.0
collected 44 items
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tests/test_train.py::test_train_model_returns_random_forest
```

```

PASSED      [      63%]      tests/test_train.py::test_train_model_is_fitted      PASSED
[      65%]      tests/test_train.py::test_train_model_prediction_length      PASSED      [
68%]      tests/test_train.py::test_train_model_predictions_are_numeric      PASSED
[      70%]      tests/test_train.py::test_train_model_predictions_positive      PASSED      [
72%]      tests/test_train.py::test_train_model_respects_n_estimators      PASSED
[      75%]      tests/test_train.py::test_train_model_respects_max_depth      PASSED
[      77%]      tests/test_train.py::test_train_model_respects_min_samples_leaf
PASSED      [      79%]      tests/test_train.py::test_train_model_reproducible      PASSED
[      81%]      tests/test_tune.py::test_tune_model_returns_dict      PASSED      [      84%]
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tests/test_tune.py::test_tune_model_best_params_within_grid      PASSED      [      90%]
tests/test_tune.py::test_tune_model_best_params_contains_all_grid_keys      PASSED
[      93%]      tests/test_tune.py::test_tune_model_best_score_is_r2      PASSED      [      95%]
tests/test_tune.py::test_tune_model_default_param_grid      PASSED      [      97%]
tests/test_tune.py::test_tune_model_best_estimator_can_predict PASSED [100%]

```

44 passed in 11.44s

(base) jovyan@a3f8dfc6013f:~/work/ML-Testing\$ python3 -m pytest tests/

-cov=firepredict

-cov-report=html

-cov-report=term-missing

-v

test session starts

platform linux -- Python 3.12.11, pytest-9.0.2, pluggy-1.6.0 -- /opt/conda/bin/python3
cachedir: .pytest_cache
rootdir: /home/jovyan/work/ML-Testing
configfile: pyproject.toml
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tests/test_evaluate.py::test_get_metrics_mae_non_negative      PASSED      [      43%]
tests/test_evaluate.py::test_get_metrics_mae_leq_mse_sqrt      PASSED      [      45%]

```

```

tests/test_evaluate.py::test_get_metrics_train_r2_geq_test_r2      PASSED      [ 47%]
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```

```

=====
tests coverage =====

```

coverage: platform linux, python 3.12.11-final-0

0.1 Name Stmtns Miss Cover Missing

```

firepredict/init.py 2 0 100% firepredict/data.py 22 0 100% firepredict/evaluate.py 10
0 100% firepredict/train.py 7 0 100% firepredict/tune.py 11 1 91% 54
----- TOTAL 52 1 98% Coverage HTML written to dir htmlcov

```

44 passed in 18.41s